

CHIME End-of-Semester Report (By Smruti Singh)

1. Project Background and Initial Direction

This project focused on understanding the relationship between climate conditions, air quality, and respiratory-related emergency visits. The overall objective was to analyze whether changes in environmental conditions such as Air Quality Index (AQI), temperature, and precipitation had a measurable impact on asthma and Chronic Obstructive Pulmonary Disease (COPD) emergency visit cases.

The project began with three major datasets:

1. Syndromic surveillance dataset containing respiratory emergency visit information.
2. Air Quality dataset (AQI) containing county-level air quality measurements.
3. ERA5 climate dataset containing weather-related variables such as temperature and precipitation.

The initial research direction was to combine these datasets at the county and date level to study how environmental conditions influence respiratory health outcomes. The core problem being addressed was whether worsening air quality and changing weather conditions could help explain increases in respiratory emergency cases.

The analysis primarily focused on:

- Asthma-related emergency visits
- COPD-related emergency visits
- Daily and monthly AQI trends
- Climate variables including temperature and precipitation
- Predictive modeling of respiratory cases using environmental variables

2. Data Cleaning and Filtering Methods

Several preprocessing and filtering techniques were applied before analysis to ensure consistency across datasets.

AQI Data Preparation

The AQI dataset was filtered to remove interpolated observations. Only records where `Was_Interpolated = FALSE` were retained in the analysis. This ensured that only directly measured AQI values were used.

County names were standardized using lowercase formatting and removal of spaces to improve merge consistency across datasets. Date columns were also converted into proper datetime format for aggregation and merging.

The AQI data was aggregated at the monthly level using mean AQI values, county-level grouping, and year/month extraction.

Syndromic Data Preparation

The syndromic surveillance dataset was processed separately for asthma cases and COPD cases. Daily case counts were aggregated by county and date. Missing values after merging were identified using null value checks.

Climate Data Preparation

The ERA5 climate dataset was used to incorporate weather variables into the final analysis. Temperature and precipitation variables were extracted and aligned with the respiratory and AQI datasets.

Feature Engineering

Several lag features were created to capture delayed environmental effects on respiratory outcomes, including AQI lag by 1 day, AQI lag by 3 days, AQI lag by 7 days, and previous respiratory case counts.

Additional temporal features included month, day, and year.

Outlier and Distribution Analysis

A Shapiro-Wilk test was applied to evaluate normality. Z-score analysis was used to identify anomalies where $|z\text{-score}| > 3$.

Data Exclusions

The following preprocessing decisions were made:

- Interpolated AQI records were excluded
- Rows with missing values after lag creation were removed
- Null records after merges were handled using drop operations
- Data was sorted by county and date before feature engineering

3. Analysis and Modeling Approach

After preprocessing, the analysis focused on identifying relationships between environmental variables and respiratory emergency cases.

Variables Used in Modeling

The final feature set included:

- Average AQI
- Average temperature
- Average precipitation
- AQI lag features
- Previous respiratory case counts
- Month feature

Machine Learning Models Used

1. Linear Regression

Linear Regression was used as the baseline model to evaluate whether a simple linear relationship existed between environmental variables and respiratory cases.

2. Random Forest Regressor

Random Forest Regression was used to capture non-linear relationships and interactions between AQI, weather conditions, and respiratory cases.

3. XGBoost Regressor

XGBoost was implemented to improve predictive performance through gradient boosting techniques.

4. Decision Tree Regressor

Decision Tree Regression was tested to evaluate rule-based partitioning of environmental variables and respiratory outcomes.

5. Gradient Boosting Regressor

Gradient Boosting Regression was implemented using 100 estimators, a learning rate of 0.1, and a maximum depth of 3.

Evaluation Metrics

The models were evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

Issues Encountered During Analysis

Challenges included:

- Missing values after merging datasets
- Loss of rows after lag feature creation
- Data imbalance across counties and time periods
- Non-normal distribution of respiratory case counts
- Outliers in emergency visit data

4. Results and Interpretation

County-Level Respiratory Case Findings

The analysis showed that respiratory emergency visits were concentrated heavily in a few Illinois counties.

The counties with the highest total respiratory emergency visits were:

County	Total Cases
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Champaign	18,153
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Macon	18,590
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McLean	10,773
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Vermilion	10,488
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These counties consistently showed higher asthma and COPD activity throughout the study period, indicating that they experienced the strongest respiratory burden among all counties included in the dataset.

Lower-volume counties included:

- Piatt
- Douglas
- Ford
- Edgar
- Moultrie
- Cumberland

These counties showed significantly fewer respiratory-related emergency visits across all years.

Seasonal Trends in Respiratory Cases

The monthly aggregation analysis revealed clear seasonal respiratory patterns.

Average respiratory case counts by month showed:

Month	Average Cases
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October	Highest (~56.9)
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January	High (~54.1)
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November	High (~52.9)
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September	High (~51.7)
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June–July	Lowest (~37–38)
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This indicates that respiratory emergency visits peaked primarily during:

- Fall (September–November)
- Winter (December–February)

while summer months experienced the lowest respiratory burden.

The seasonality plots also showed that respiratory cases increased during periods where AQI values fluctuated significantly.

Model Performance Results

The machine learning comparison produced the following MAE scores:

Model	MAE
Linear Regression	1.556
XGBoost	1.624
Random Forest	1.659
Decision Tree	2.206

Key interpretation:

- Linear Regression surprisingly performed best on this dataset.
- Decision Tree performed worst due to overfitting and instability.

- Ensemble models captured more complex relationships but did not drastically outperform linear modeling.
- The relatively low MAE values indicate that environmental variables contributed meaningfully to predicting respiratory cases.

Major findings:

- Macon and Champaign counties experienced the highest respiratory burden.
- Respiratory cases peaked primarily during fall and winter.
- AQI peaked during warmer months but influenced later respiratory outcomes.
- Lagged AQI exposure improved predictive performance.